Queuing Theory: Part 1

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Queuing Theory

- Queuing theory is a branch of Operations Research that studies how waiting **lines** form, how they function, and why they malfunction.
- The word "**line**" does not only mean a physical line of people. It refers more broadly to any situation where **customers** **(people, data, jobs) wait for service**
- The origin of queuing theory can be traced to the early 1900s in a study of the Copenhagen (city in Denmark) telephone exchange by Agner Krarup Erlang, a Danish engineer, statistician, and mathematician
- Queuing theory studies behavior of waiting lines (queues)
  - It examines every component of waiting lines, including the arrival process and the number of customers among others.

Queuing Theory

- Models the arrival, service, waiting, and departure of entities.
- Uses probability and mathematical tools.
- Queuing theory aims to achieve a balance that is efficient and affordable.
- Used as an operations management tool, queuing theory can address scheduling, and customer service shortfalls
- At a basic level, queuing theory involves the analysis of arrivals at a facility, such as a bank or a fast-food restaurant. It analyses the processes currently in place to serve them.

Real life examples of queues

Location	Customers	Service facilities
Petrol pump	Vehicles	Fuel dispenser
Hospital	Patients	Doctors, Nurses, Beds etc.
Job Interview	Candidates	Interviewers
Railway station	Traveller	Ticket counter, Trains
Service centre	Cars	Washing facility, mechanic

Arrangement of service facilities

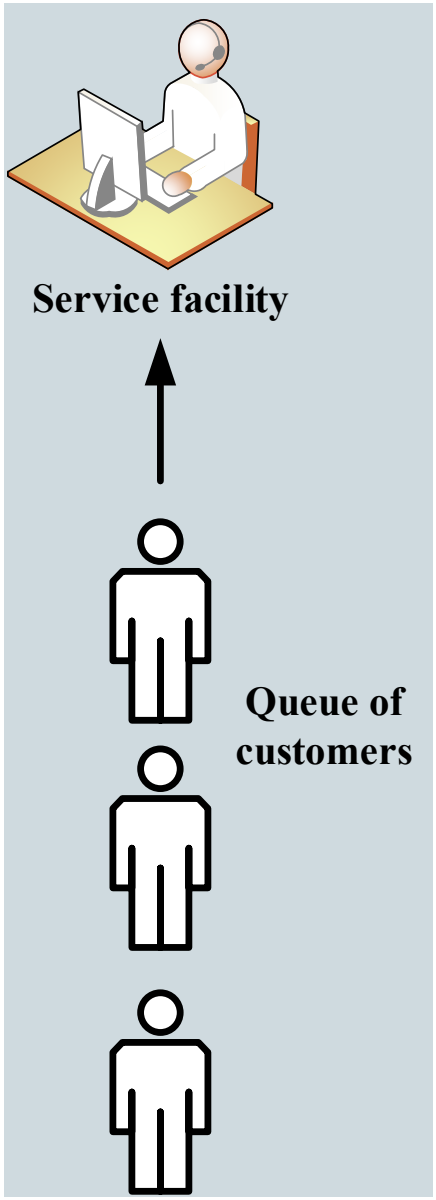


Figure Single queue, single facility

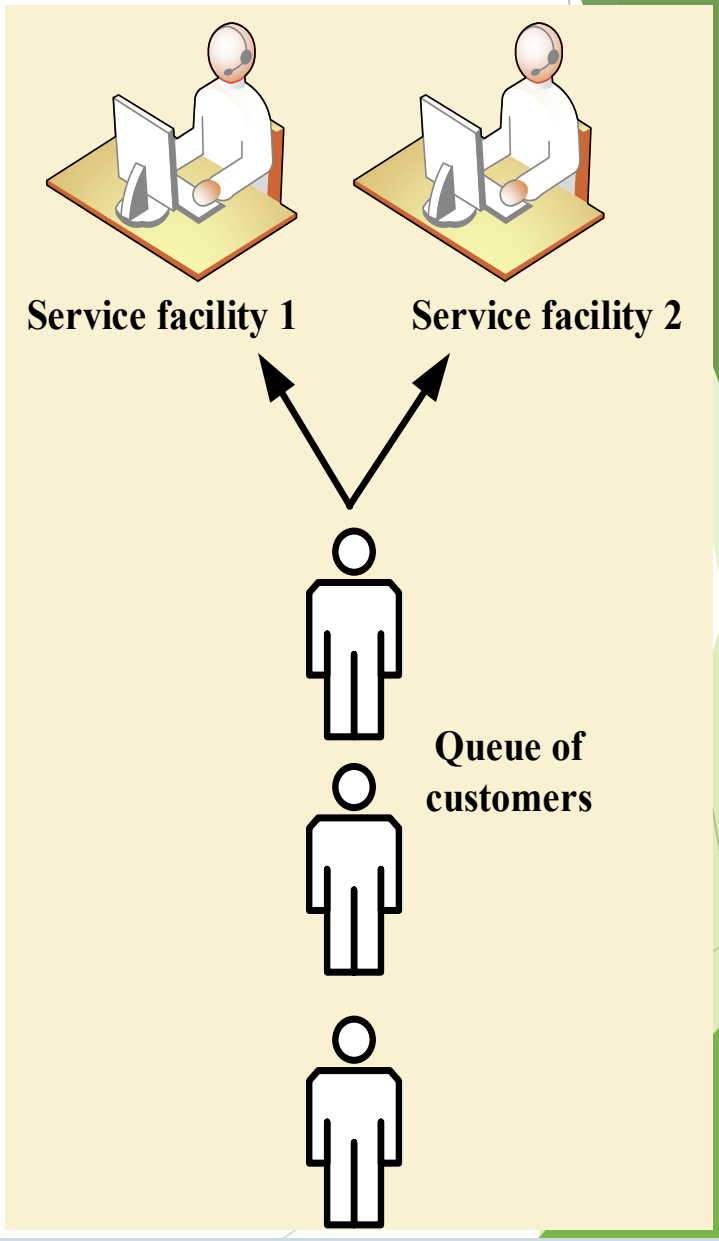


Figure Single queue, multiple facility

Arrangement of service facilities

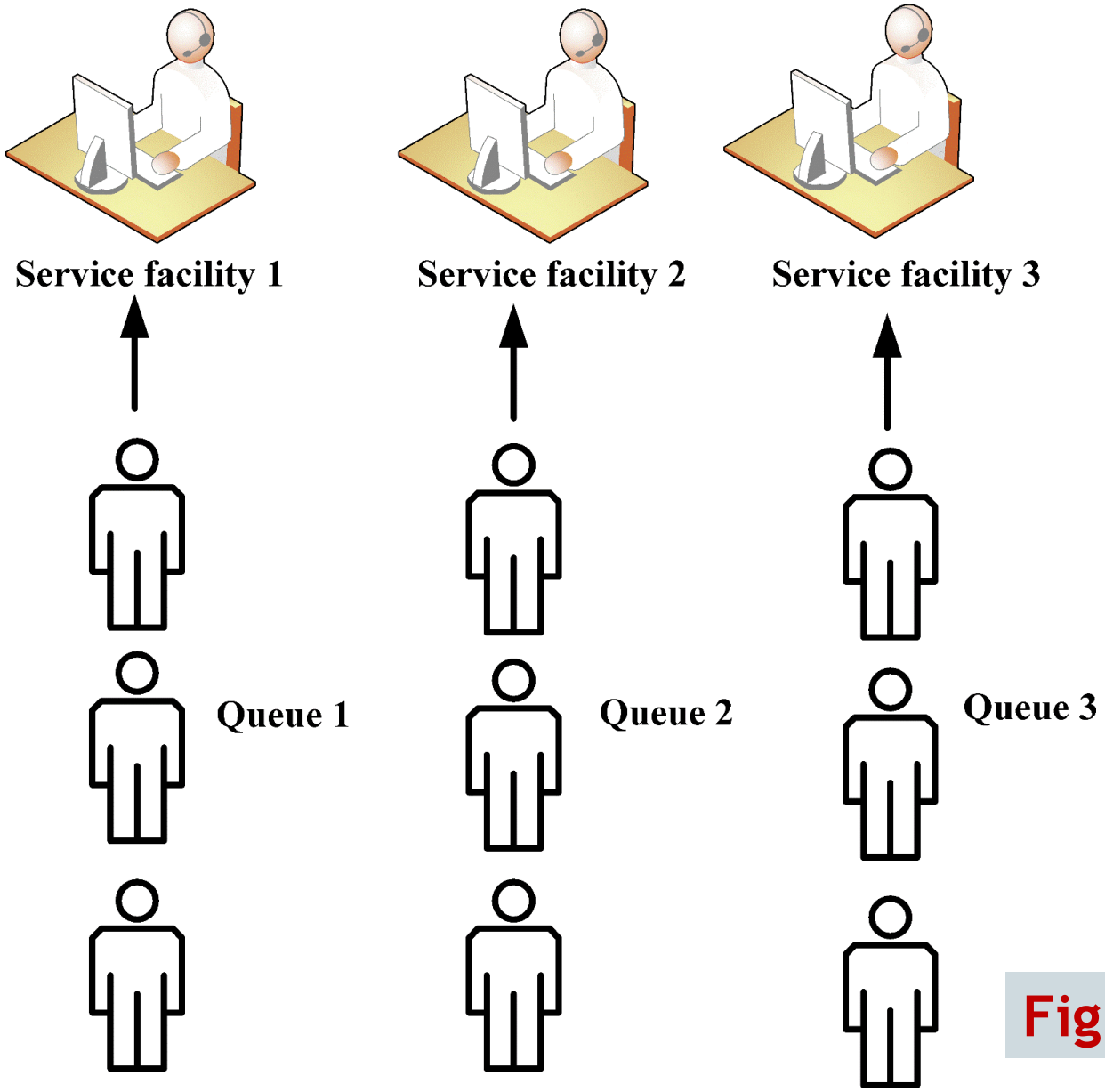


Figure Multiple queues, multiple facilities

Elements of a queuing model

- Main players in a queuing situation are **customer** and **server**.
- Customers arrive at the service (facility) from a source. On arrival, the service may start immediately or the customer may wait in a **queue** if the facility is busy.
- When the service is completed, it pulls a waiting customer from the queue. If the queue is empty, the facility becomes idle until a new customer arrives.
- The arrival of customers is represented by **interarrival time** (*time between successive arrivals*).
- The service is measured by the **service time** per customer.

Elements of a queuing model

- **Queue discipline** represents the order in which customers are selected from a queue. Common disciplines are FIFO, LIFO and SIRO.
  - **First-in, first-out (FIFO):** The first item or person to enter a system is the first one to be processed or served. Examples:

    Printer jobs: The first document sent to the printer gets printed before the next one.

    Queues at a ticket counter: The person who arrives first gets served first.

    Inventory management: The oldest stock (first in) is sold or used first in grocery stores to avoid spoilage.

Elements of a queuing model

- **Last-in, first-out (LIFO):** The last item to enter is the first one to leave. Examples:
  - **Undo function in software:** Last action is the first to be undone.
  - **Stack of plates:** When plates are stacked, last plate placed (on top) is taken first.
  - **Browser history or call logs:** The last site we visited or call we made appears first.
- **Service in random order (SIRO):** **It** refers to a **queuing discipline** where customers or items are served in **no specific order**, but rather **randomly selected** from the queue.

  Examples:
  - **Resource allocation during emergencies:** When order isn't critical, resources might be distributed randomly (relief kits in some disaster zones).
  - **Lottery based prize distribution:** Tickets are drawn at random and prize winner is declared.

Elements of a queuing model

- **Queuing behavior in waiting-line analysis:**

- **Jockeying: Switching Queues**

  To reduce waiting time, customers may **jockey** from a longer queue to a shorter one to reduce waiting time, eg., at **toll booths**, drivers often change lanes hoping to get through faster.

- **Balking: Refusing to Join**

  Customers can **balk** from joining a queue altogether because of anticipated long delay, eg.: at a **restaurant**, seeing a long waiting line, someone might decide not to wait and leave.

Elements of a queuing model

- **Queuing behavior in waiting-line analysis**
- **Reneging: Leaving After Joining**

  Customers may **renege** from a queue because they have been waiting too long, eg., at a **bank**, someone takes a token, waits for a while, then leaves before their number is called.

Elements of a queuing model

- The source from which customers are generated may be **finite** or **infinite**.
- **Infinite Source:** **Number of potential customers is very large or unlimited**. Hence, the arrival of one customer does **not affect the chances** of others arriving, eg., calls arriving at a telephone exchange.
- **Finite Source:** **Total number of potential customers is limited**. When one enters the system, it **reduces the number of potential arrivals** temporarily. eg., **Machines in a factory** waiting for repair. If 10 machines are there and 3 are being repaired, only 7 machines enter the queue.

Role of exponential distribution in queuing theory

- In most queuing situations, arrivals occur randomly.
- Randomness means that the occurrence of an event (arrival of customer or completion of a service) is not influenced by the length of time that has elapsed since the occurrence of last event.
- Random inter-arrival and service times are described quantitatively in queuing models by **exponential distribution**.

- **Exponential distribution**

If the number of arrivals at a service facility during a specified time period follows the Poisson distribution, then, the distribution of inter-arrival time (between successive arrivals) is the negative exponential (or exponential) distribution.

If λ is the rate of arrival, then the probability density function of inter-arrival time, x is

$$f(x) = \lambda e^{-\lambda x}, x > 0$$

Exponential distribution

The mean and variance of exponential distribution are

$$E\{x\} = \frac{1}{\lambda}$$

$$\text{var}\{x\} = \frac{1}{\lambda^2}$$

The mean $E\{x\}$ is consistent with the definition of λ.

The λ is the rate at which events occur; $1/\lambda$ is the average time interval between successive events.

Key difference between Poisson and exponential distribution

Poisson Distribution	Exponential Distribution
Number of events in time	Time between events

Example: A customer care receives call randomly. The average inter-arrival time is 3 minutes. What is the probability that the inter-arrival time does not exceed 2 minutes?

Solution: Determining the probability is same as computing the cumulative distribution function (CDF) of x.

$$P\{x \le t\} = \int_0^t \lambda e^{-\lambda x} \mathrm{d}x$$

$$= -e^{-\lambda x}\Big|_o^t$$

$$= 1 - e^{-\lambda t}$$

The arrival rate is λ = 1/3 arrival per minute. Substituting t = 2, the probability is obtained as

$$P\{x \le 2\} = 1 - e^{-\left(\frac{1}{3}\right)(2)} = 0.487$$

More on probability density function

$$f(x) = \lambda e^{-\lambda x}, x > 0$$

The probability of x lying between 0 and ∞ is 1.

This is true, because

$$\int_0^\infty f(x)\,\mathrm{d}x = \int_0^\infty \lambda e^{-\lambda x}\,\mathrm{d}x = \left[-e^{-\lambda x}\right]_0^\infty = 1.$$

Expected value (mean) of exponential distribution

The **expected value** $\mathbb{E}[X]$ of a continuous random variable X is given by:

$$\mathbb{E}[X] = \int_{-\infty}^{\infty} x f(x)\, dx$$

Since $f(x) = 0$ for $x < 0$, we only integrate from 0 to ∞:

$$\mathbb{E}[X] = \int_{0}^{\infty} x\lambda e^{-\lambda x}\, dx$$

$$\mathbb{E}[X] = \left[-xe^{-\lambda x}\right]_0^{\infty} + \int_0^{\infty} e^{-\lambda x} dx$$

$$\mathbb{E}[X] = 0 + \int_0^{\infty} e^{-\lambda x} dx = \left[-\frac{1}{\lambda} e^{-\lambda x}\right]_0^{\infty} = 0 + \frac{1}{\lambda}$$

Variance of exponential distribution

$$\mathrm{Var}(X) = \int_0^\infty (x - \bar{x})^2 \cdot f(x)\, dx = \int_0^\infty \left(x - \frac{1}{\lambda}\right)^2 \cdot \lambda e^{-\lambda x}\, dx$$

$$= \int_0^\infty \left(x^2 - \frac{2x}{\lambda} + \frac{1}{\lambda^2}\right) \cdot \lambda e^{-\lambda x}\, dx$$

$$= \lambda \int_0^\infty x^2 e^{-\lambda x} dx - \frac{2}{\lambda}\lambda \int_0^\infty x e^{-\lambda x} dx + \frac{1}{\lambda^2}\lambda \int_0^\infty e^{-\lambda x} dx$$

$$\mathrm{Var}(X) = \lambda \cdot \frac{2}{\lambda^3} - \frac{2}{\lambda} \cdot \lambda \cdot \frac{1}{\lambda^2} + \frac{1}{\lambda^2} \cdot \lambda \cdot \frac{1}{\lambda}$$

$$= \frac{2}{\lambda^2} - \frac{2}{\lambda^2} + \frac{1}{\lambda^2} = \boxed{\frac{1}{\lambda^2}}$$